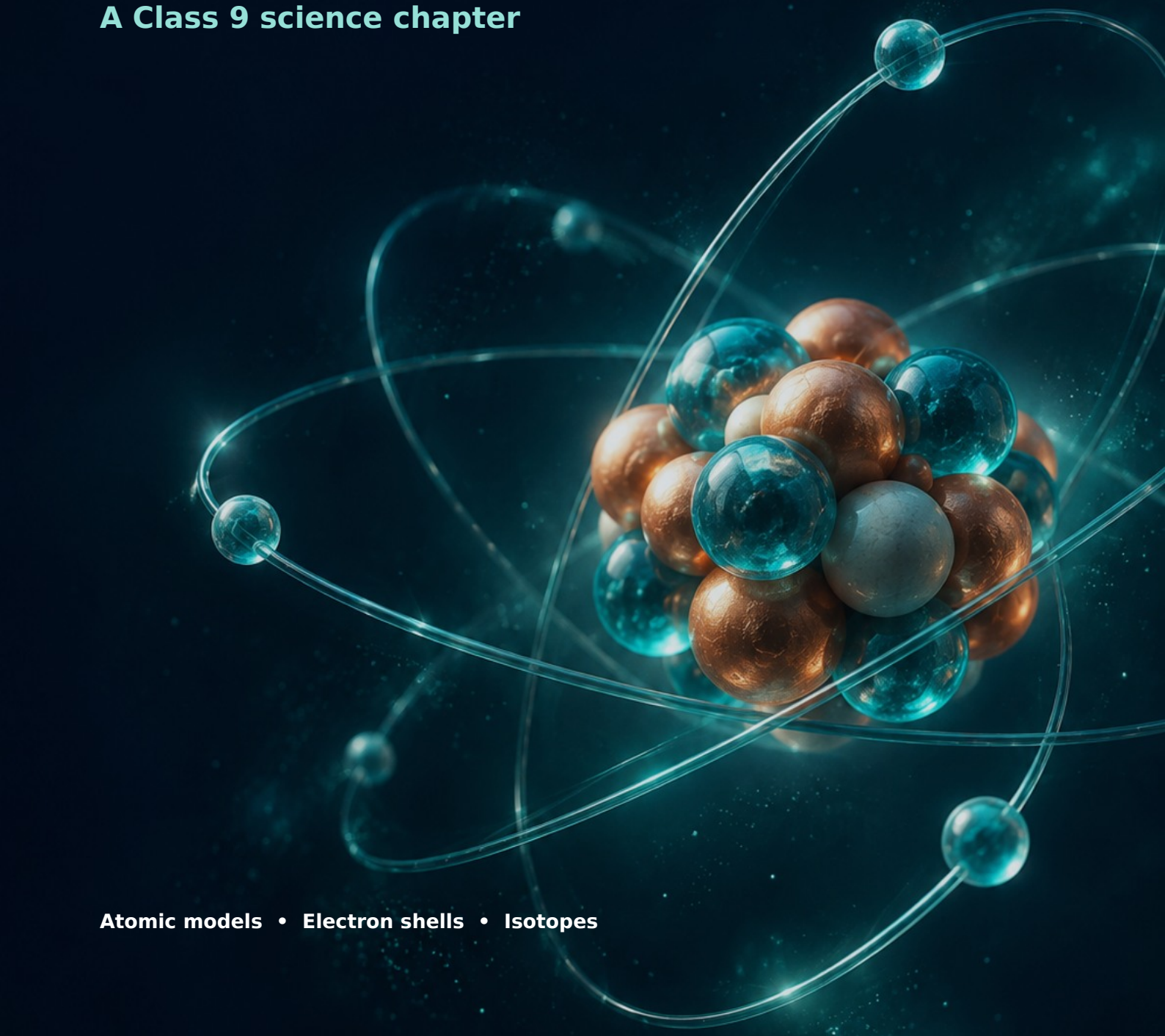
RAMA’S ACADEMY

CLASS 9 / SCIENCE

Journey Inside the Atom

A Class 9 science chapter



Atomic models • Electron shells • Isotopes

CLASS 9 SCIENCE

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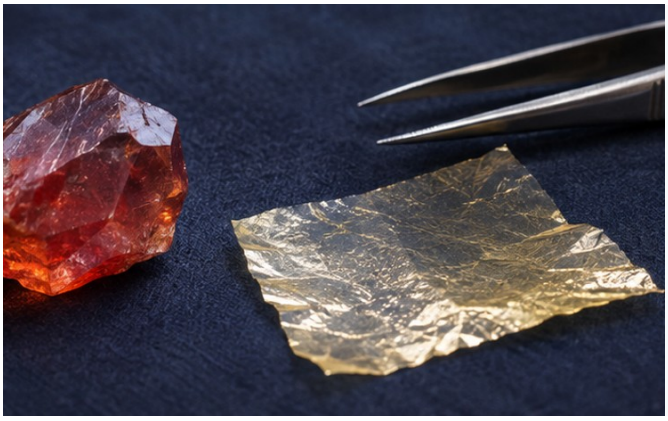
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I / EVIDENCE FOR THE ATOM

A question small enough to change everything

A strip of gold leaf seems perfectly smooth. Tear it and each piece is still gold. Keep imagining smaller and smaller pieces. At some point, could you reach a piece that still counts as gold but has no smaller piece that is itself gold? That question takes us from an ordinary object to the idea of an atom, the smallest unit of an element that retains its chemical identity.



From the visible to the atomic

Foil and a mineral specimen are visible samples. Their atoms are far too small to see unaided.

An idea becomes a scientific model

In ancient India, Acharya Kanada described tiny units of matter called **parmanu**. In ancient Greece, Leucippus and Democritus also reasoned that matter might be made of very small particles. Those ideas were philosophical. In the early nineteenth century, John Dalton used measurements from chemical reactions to give atomic theory a scientific footing.

Dalton’s model explained why elements combine in simple whole-number ratios. If a substance contains two hydrogen atoms for each oxygen atom, adding more of that substance adds more of both atoms in the same ratio. Dalton treated atoms as indivisible and pictured the atoms of one element as identical in mass. The next discoveries would show that atoms have parts, and that atoms of one element can have different masses.

A million atoms across

A sheet is 0.10 mm thick. If an atom is about 10^{-10} m across, estimate how many atomic diameters fit across the thickness.

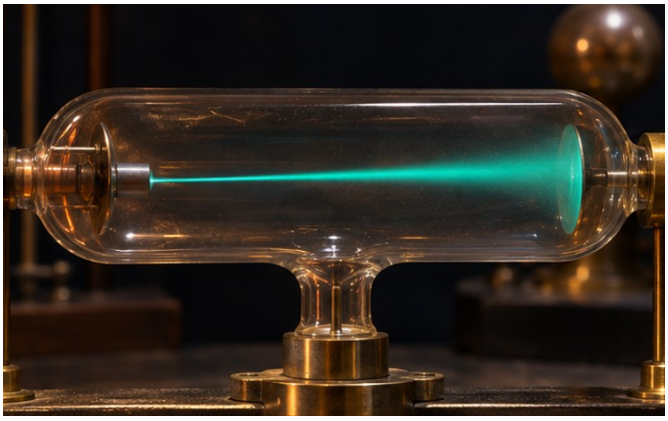
1. Convert the thickness first. 0.10 mm = 1.0×10^{-4} m.

$$N \approx \frac{10^{-4}\,\text{m}}{10^{-10}\,\text{m}} = 10^{6}$$

2. About one million atomic diameters fit across. This is a scale estimate, since actual atoms and the spacing between them vary.

A glowing tube reveals the electron

If an atom really had no smaller parts, there should be no way to draw identical tiny charged particles from different materials. Experiments challenged that idea. Radioactivity showed that certain atoms could emit particles or radiation. J. J. Thomson then investigated a beam that formed when a high voltage was applied across electrodes in a low-pressure glass tube.

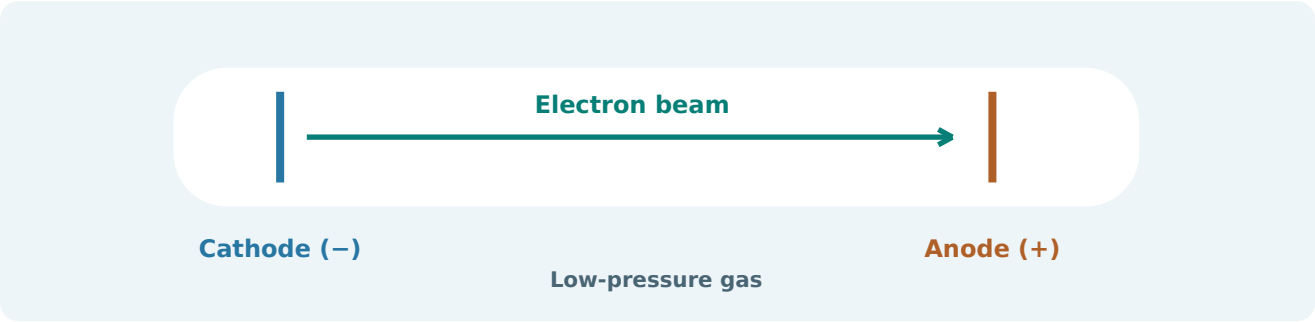


A discharge tube

A glow in a low-pressure gas can make parts of the discharge visible. The diagram below shows the direction of the electron beam.

How the experiment worked

The negative electrode is the **cathode**, and the positive electrode is the **anode**. The beam, called a cathode ray, travelled away from the cathode. Electric and magnetic fields bent its path. Its deflection towards a positive plate showed that the particles in the beam carried negative charge. Measurements also showed that the particles were far lighter than atoms.

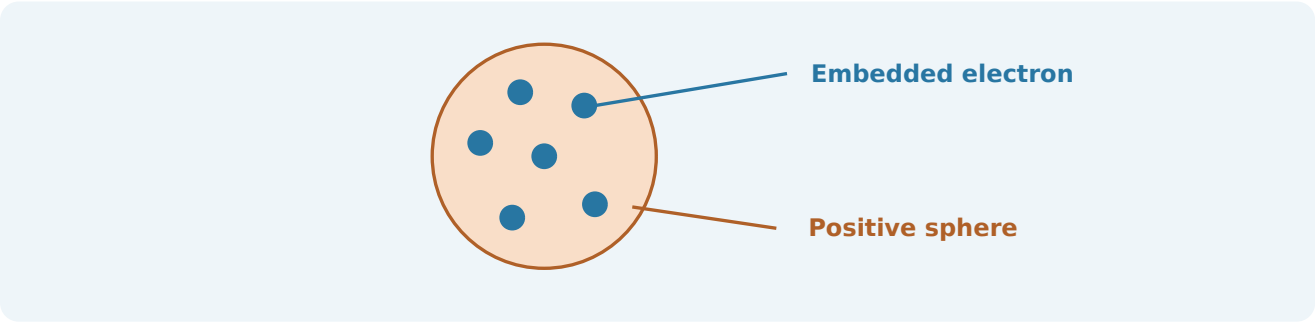


A simplified cathode-ray tube. Electrons travel away from the negative electrode.

When Thomson changed the gas or the electrode material, the beam still behaved as a stream of the same kind of particle. That made sense if the **electron** was a component shared by atoms of different elements. We represent it as e^- and assign it relative charge −1. This created the next puzzle. Ordinary atoms are electrically neutral, so where was the positive charge that balanced the electrons?

From Thomson’s model to the gold-foil test

Thomson’s answer was to spread the positive charge throughout a sphere and embed electrons inside it. Think of seeds embedded in the flesh of a watermelon. In this model, positive and negative charges add to zero when the atom is neutral. It explained how an atom could contain electrons and still have no net charge.



Thomson placed the electrons within a continuous positive sphere.

A prediction worth testing

Ernest Rutherford and his colleagues Hans Geiger and Ernest Marsden used a narrow beam of positively charged **alpha particles** to probe a thin sheet of gold. A surrounding fluorescent screen revealed where particles arrived by producing tiny flashes. Alpha particles are helium nuclei, with charge +2 in relative units. Gold can be beaten into very thin foil, so the beam could probe a thin layer of atoms.



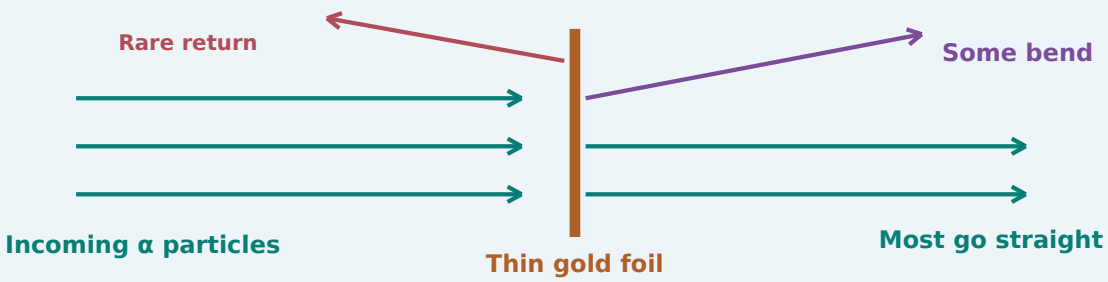
A thin gold target

Gold can be made into very thin leaf. The scattering experiment used thin foil so most particles passed through.

If the atom’s positive charge really were spread out, a fast alpha particle passing through the foil would encounter gentle pushes from diffuse positive charge. The investigators expected most particles to pass through with little or no deflection. The result that mattered most would come from the few particles that did something different.

The rare bounce that revealed the nucleus

Most alpha particles went straight through the foil, some changed direction, and a very small number scattered through large angles, including backwards. **Scattering** means a change in direction. A large deflection is a clue that the moving particle experienced a strong electric force during a close encounter.



Particle paths are schematic. A detector screen records where particles emerge.

Link each observation to an inference

What the screen showed	What the observation implies
Most particles went nearly straight	Most paths miss a strongly repelling region. The nucleus occupies only a tiny fraction of the atom.
Some particles bent away	A passing positive alpha particle can be repelled by concentrated positive charge.
Very few particles returned towards the source	Rare close approaches to a small, dense, positively charged nucleus can cause large deflections.

Rutherford's nuclear model concentrated positive charge and almost all the mass in a tiny **nucleus**. The electrons occupied the much larger region around it. The large-angle events were hard to explain with Thomson's diffuse positive sphere. They made sense once the positive charge was concentrated in a small region. A typical atom is of order 10^{-10} m across, while nuclear dimensions are of order 10^{-15} m.

What if the foil is thicker?

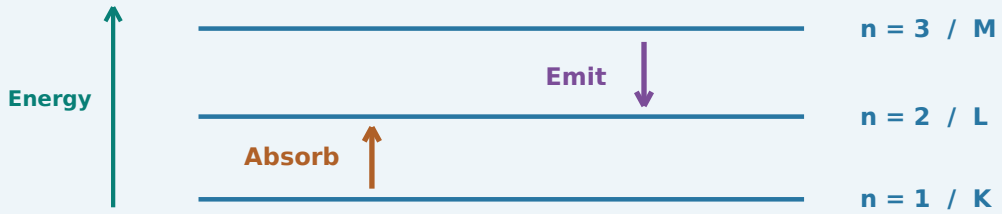
1. Keeping the beam and material the same, a thicker foil gives particles more opportunities to pass near nuclei. More particles can be scattered, sometimes more than once. The fraction going straight through is expected to fall.

Why electrons need energy levels

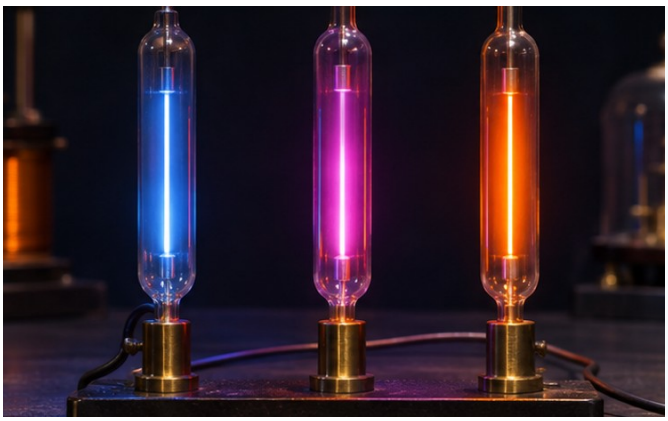
Rutherford pictured electrons revolving around the nucleus. The model explained the scattering, but classical physics predicted a serious problem. An electron changing direction in a circular orbit is accelerating. According to classical physics, an accelerating charge radiates energy. A continuously radiating electron would lose energy and spiral inward, yet atoms remain stable.

Bohr's new rule

Niels Bohr proposed that only particular energies were allowed. In his model, an electron in one of these **stationary states** keeps its energy. The allowed levels are labelled n = 1, 2, 3, …, or K, L, M, … . The level nearest the nucleus is the lowest in energy in this simple model. An electron can absorb energy and move to a higher level, or emit energy as light when it drops to a lower one.



An upward transition takes in energy. A downward transition releases energy as light.



Light from excited gases

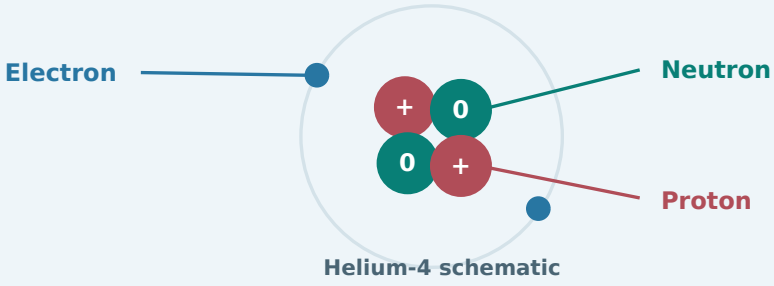
Different gases produce characteristic spectral lines when excited. These glowing tubes illustrate the kind of light that can be studied.

Different energy gaps produce different frequencies of light. This helps explain why an excited gas produces distinct spectral lines. Bohr’s model works particularly well for hydrogen and gives a useful shell picture for Class 9. In the modern model, electrons are described by probability distributions around the nucleus. The rings in this chapter are energy-level diagrams and do not show measured electron tracks.

II / READING AN ATOM

The missing mass in the nucleus

A new problem remained. A helium nucleus has two protons but a mass close to four proton masses. Its two electrons are far too light to account for the difference. In 1932, James Chadwick identified the **neutron**, an electrically neutral particle with a mass close to a proton’s mass. Two protons and two neutrons explain the main contribution to the mass of a helium-4 atom.



Helium-4 contains two protons, two neutrons and, when neutral, two electrons.

Particle	Relative charge	Approximate mass	Location
Electron, e^-	−1	About 1/1836 u	Region around the nucleus
Proton, p^+	+1	About 1 u	Nucleus
Neutron, n^0	0	About 1 u	Nucleus

Protons and neutrons are collectively called **nucleons**. Hydrogen-1 has a single proton and no neutron. Other hydrogen isotopes do have neutrons. An atom is neutral when its proton and electron counts match. Neutrons change nuclear mass without changing electric charge. The short-range nuclear force helps bind nucleons despite the repulsion between protons.

Science in India

Homi J. Bhabha helped establish India’s atomic research programme and institutions including TIFR and the Atomic Energy Establishment, Trombay, later named BARC. At BARC, neutron scattering helps researchers study how atoms are arranged and move in materials. A beam of neutrons provides evidence about a material’s structure in much the same spirit as the gold-foil experiment used scattering to investigate the atom.

Reading the name, number and mass of an atom

The symbol tells us the element. Dalton used pictorial symbols, while Berzelius helped establish the letter-based system behind modern chemical symbols. Start a symbol with a capital letter. If there is a second letter, it is lowercase. For example, Co is cobalt, while CO is the formula for carbon monoxide. Some symbols come from older names, including Na for sodium, Fe for iron, and K for potassium.

Atomic number and mass number

The **atomic number** Z is the number of protons. Change Z and the element changes. The **mass number** A is the number of protons plus neutrons in a particular atom. It is a whole-number count with no unit. In nuclear notation, A goes above and to the left of the symbol and Z goes below and to the left.



Read the mass number above the atomic number. Both are written to the left of the symbol.

$$Z = N_{\text{protons}} \quad A = Z + N_{\text{neutrons}} \quad N_{\text{neutrons}} = A - Z$$

Read aluminium-27

For neutral aluminium-27, Z = 13 and A = 27. Find the number of protons, electrons and neutrons.

1. Atomic number gives 13 protons. A neutral atom has the same number of electrons, so there are 13 electrons.

$$N_{\text{neutrons}} = 27 - 13 = 14$$

2. Check the mass number by adding 13 protons and 14 neutrons. The result is 27.

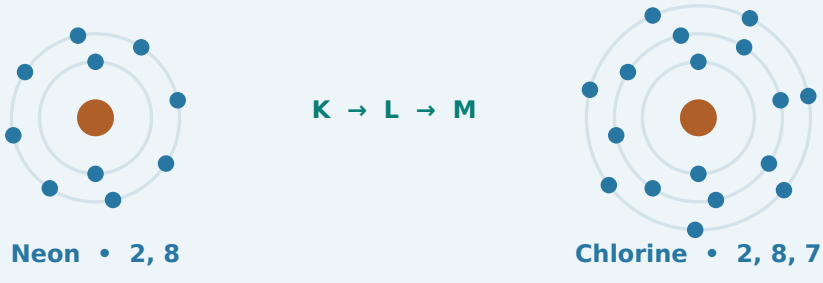
Filling the shells from the inside out

Having counted an atom’s electrons, we can ask how they are distributed. The **electronic configuration** lists the number in each shell. The first shells are K, L, M and N, corresponding to n = 1, 2, 3 and 4. The maximum number of electrons a shell can hold is given by the rule below. A maximum capacity is different from the number actually present.

$$\text{Maximum shell capacity} = 2n^2$$

Shell	K	L	M	N
Shell number n	1	2	3	4
Maximum capacity	2	8	18	32

For the first 18 elements, use this school-level pattern. Fill K up to 2, then L up to 8, then M up to 8 by argon. The maximum capacity of M is 18, but its eight electrons in argon are enough to fill the outer-shell pattern used here. In higher elements, subshell energies determine the filling order. Potassium and calcium, for example, begin filling N even though M has fewer than 18 electrons.



Count the blue electron markers from the inner shell outwards.

Build the configuration of chlorine

1. Chlorine has Z = 17, so a neutral chlorine atom has 17 electrons. Put 2 in K and 8 in L.
2. The remaining 17 − 2 − 8 = 7 electrons go in M. The configuration is 2, 8, 7. Its three entries add to 17.

The first eighteen elements

The table connects three ideas. The atomic number gives the proton count and, for a neutral atom, the electron count. The configuration tells us how those electrons are distributed. Each mass number A in the table identifies one isotope of that element. The final column gives the simple valency commonly used at this level. Some elements, such as sulfur and phosphorus, can show other valencies in different compounds.

Element	Symbol	Z	A	p^+	n^0	K, L, M configuration	Valency
Hydrogen	H	1	1	1	0	1	1
Helium	He	2	4	2	2	2	0
Lithium	Li	3	7	3	4	2, 1	1
Beryllium	Be	4	9	4	5	2, 2	2
Boron	B	5	11	5	6	2, 3	3
Carbon	C	6	12	6	6	2, 4	4
Nitrogen	N	7	14	7	7	2, 5	3
Oxygen	O	8	16	8	8	2, 6	2
Fluorine	F	9	19	9	10	2, 7	1
Neon	Ne	10	20	10	10	2, 8	0
Sodium	Na	11	23	11	12	2, 8, 1	1
Magnesium	Mg	12	24	12	12	2, 8, 2	2
Aluminium	Al	13	27	13	14	2, 8, 3	3
Silicon	Si	14	28	14	14	2, 8, 4	4
Phosphorus	P	15	31	15	16	2, 8, 5	3
Sulfur	S	16	32	16	16	2, 8, 6	2
Chlorine	Cl	17	35	17	18	2, 8, 7	1
Argon	Ar	18	40	18	22	2, 8, 8	0

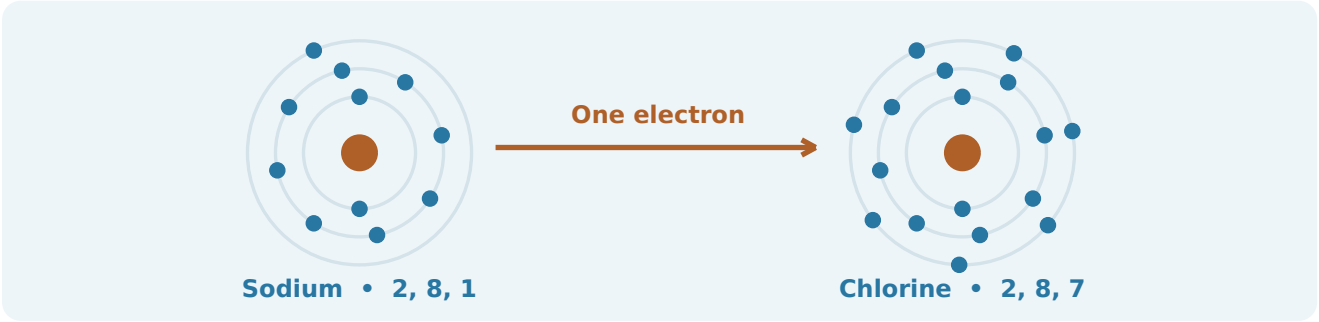
The next atom after neon

1. Neon has 10 electrons and configuration 2, 8. Sodium has 11 electrons, so the next electron starts a new shell.
2. Sodium is 2, 8, 1. That one electron in its outer shell makes its chemical behaviour very different from neon’s, even though their atomic numbers differ by only one.

Why the outer shell matters

The electrons in the outermost occupied shell are the **valence electrons**. They are the ones most directly involved when atoms form bonds. The **valency** is an atom’s combining capacity. In many simple compounds, you can work it out by counting how many electrons are lost, gained or shared as the atom reaches a stable arrangement.

Helium has a full K shell with two electrons, called a **duet**. Neon has eight electrons in its outer L shell, called an **octet**. These noble gases are largely unreactive and are assigned valency zero at this level. Sodium can lose its one outer electron to leave the arrangement 2, 8. Chlorine can gain one electron to reach 2, 8, 8. Each has valency 1. Oxygen has six valence electrons and commonly has valency 2. Carbon has four valence electrons and typically shares electrons to form four bonds.



The transfer picture gives Na^{+} and Cl^{-}. Their charges come from the change in electron count.

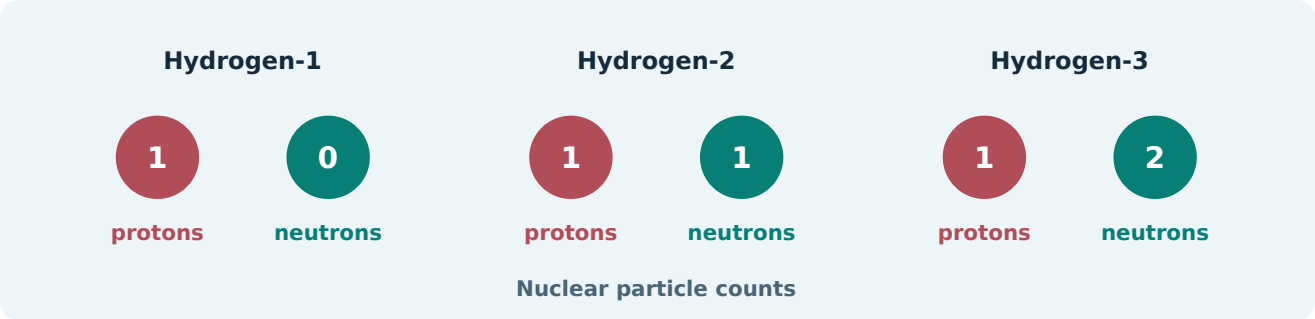
Three ways to use the outer-shell count

1. Magnesium is 2, 8, 2. It can lose two outer electrons, so its common valency is 2.
2. Oxygen is 2, 6. It needs two more electrons for an octet, so its common valency is 2.
3. Carbon is 2, 4. It commonly shares electrons in four bonds, so its valency is 4. Valency is written as a number without a charge sign. Ion charge, by contrast, carries a sign.

III / SAME ELEMENT, DIFFERENT MASS

Same element, different mass

Dalton’s model treated all atoms of an element as identical in mass. Atomic number now gives a more precise test of identity. All hydrogen atoms contain one proton, but the nucleus can contain zero, one or two neutrons. These are hydrogen-1 (protium), hydrogen-2 (deuterium) and hydrogen-3 (tritium). They are **isotopes**: the same Z, different A.



The proton count stays at one while the neutron count changes.

Carbon-12, carbon-13 and carbon-14 all have six protons, but have six, seven and eight neutrons respectively. The neutral atoms have the same electron configuration, so their chemistry is generally similar. Their masses and nuclear behaviour can differ. Carbon-14 and tritium are radioactive. Some isotopes are stable.

From isotope to application

Uranium-235 can undergo fission and is used in nuclear fuel. Cobalt-60 emits penetrating radiation and has applications in radiotherapy and sterilisation. Iodine-131 has controlled medical uses for certain thyroid conditions. These applications depend on each isotope’s nuclear behaviour. Carbon-14 decay can help date once-living material.

Isotope identity check

Two neutral atoms have 17 protons each. The first has 18 neutrons and the second has 20.

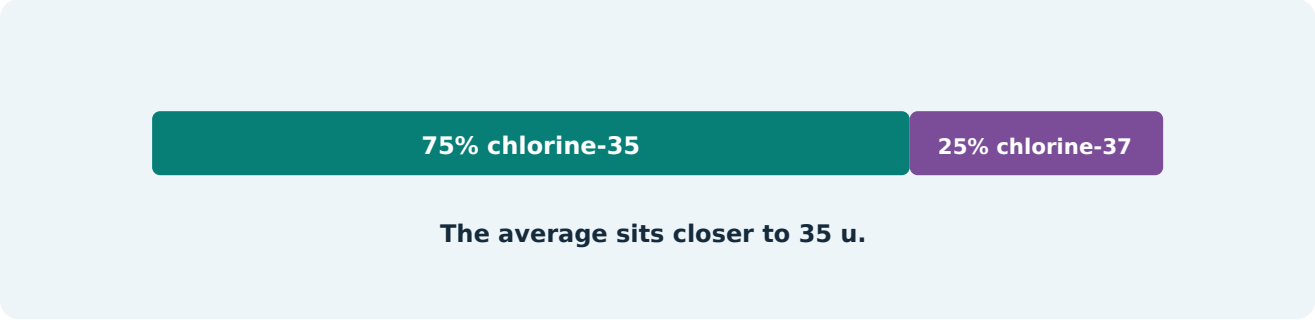
1. Both are chlorine because Z = 17. Their mass numbers are 35 and 37. They are isotopes of chlorine.

Why some atomic masses have decimals

The unified atomic mass unit, u, is defined as one-twelfth of the mass of a carbon-12 atom. The mass of a particular isotope is measured in u. Its numerical value is often close to its mass number, but mass number is a count and isotope mass is a measurement. A periodic-table atomic mass usually reflects the mixture of isotopes in a natural sample.

$$\overline{m} = \sum_i f_i m_i$$

Here f is the fractional abundance and m is the isotope mass in u. Convert percentages to fractions by dividing by 100. The fractions must add to one. An isotope present in a larger fraction pulls the average closer to its mass.



The lengths represent fractional abundances in the worked chlorine example.

Average atomic mass of a chlorine sample

Assume 75% chlorine-35 and 25% chlorine-37. Use 35 u and 37 u as approximate isotope masses.

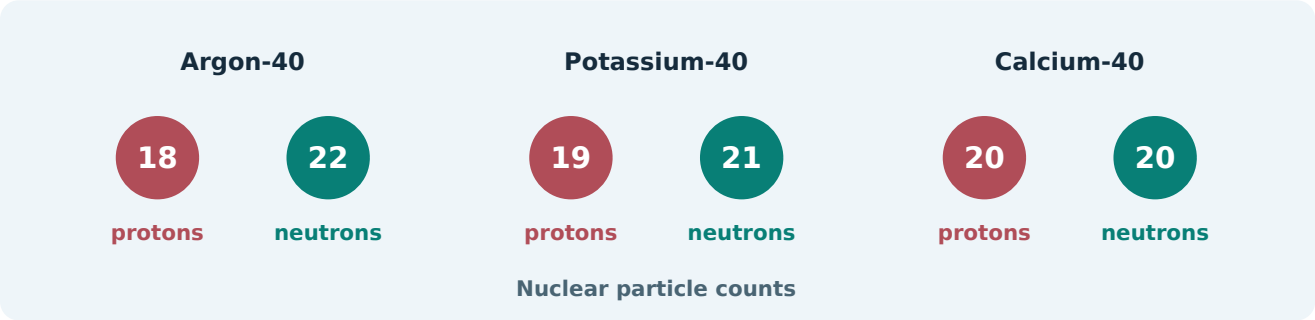
1. Convert 75% to 0.75 and 25% to 0.25. Multiply each isotope mass by its fraction and add.

$$\overline{m} = (0.75)(35\,\text{u}) + (0.25)(37\,\text{u}) = 35.5\,\text{u}$$

2. Check that 35.5 u lies between 35 u and 37 u and closer to 35 u, the more abundant isotope.

The same mass number can hide different elements

Now reverse the comparison. Argon-40, potassium-40 and calcium-40 have the same mass number, but different atomic numbers. They are **isobars**. Their total nucleon count is the same. The balance of protons and neutrons differs, so they belong to different elements.



Each nucleus contains 40 nucleons. The proton count is different in each case.

Compare	Isotopes	Isobars
Atomic number Z	Same	Different
Mass number A	Different	Same
Example	Chlorine-35 and chlorine-37	Argon-40 and calcium-40

A model changes when evidence demands it

Dalton’s atoms helped explain chemical combinations. Thomson’s beam revealed electrons. Scattering revealed the nucleus. Bohr introduced allowed energies, and Chadwick identified the neutron. Modern quantum models describe electron probability rather than fixed tracks. Tools such as scanning tunnelling microscopes can map surfaces at atomic scales, and transmission electron microscopes can reveal atomic arrangements in very thin specimens.

The gold leaf that started our journey still has the chemical identity of gold because its atoms have 79 protons. What has changed is the detail we can explain. We now have ways to talk about the nucleus, the electrons that govern much of its chemistry, and the different evidence needed to learn about each.

IV / PRACTICE AND REVIEW

Practice • Evidence and models

1. Name the particle

A beam bends towards a positive plate. What is the sign of the beam particles? Which particle did Thomson identify?

2. Read the scattering results

State what each of these observations tells us: most alpha particles passed straight through, and a few were deflected through large angles.

3. Explain the stability problem

Why would a classical electron in a circular orbit around a nucleus lose energy? How did Bohr alter the model?

4. Infer an energy change

An electron moves from n = 3 to n = 2. Does the atom absorb or emit energy? Explain.

5. Identify the missing particle

A neutral atom has two protons and two electrons, but its mass is close to 4 u. How many neutrons would you expect and why?

6. Compare length scales

An atom is approximately 10^{-10} m across and its nucleus approximately 10^{-15} m across. How many times larger is the atomic diameter?

Practice • Counts, shells and isotopes

7. Decode an atom

A neutral atom has Z = 16 and A = 34. Name the element and find its protons, neutrons, electrons and shell configuration.

8. Explain the difference

Silicon has configuration 2, 8, 4. Chlorine has configuration 2, 8, 7. Compare their valence electrons and simple valencies.

9. Classify a pair

One atom has 18 protons and 22 neutrons. Another has 20 protons and 20 neutrons. Are these isotopes or isobars?

10. Calculate an average

A sample of bromine is 49.7% bromine-79 and 50.3% bromine-81. Using mass numbers as approximate masses in u, calculate its average atomic mass.

11. Recover the abundance

A sample contains only isotopes of masses 10 u and 11 u. Its average mass is 10.8 u. Find the percentage of the heavier isotope.

12. Distinguish the numbers

Someone reads chlorine’s average atomic mass of about 35.5 u and claims that one chlorine atom has 35.5 nucleons. Explain the error.

Worked solutions • Evidence and models

1. Name the particle

1 A positive plate attracts the beam. Opposite charges attract, so the beam particles have negative charge. Thomson identified the electron.

2. Read the scattering results

1 Most paths passed through without a close encounter, so the strongly repelling region occupies a very small part of each atom.

2 A large deflection needs a strong electric force. A close approach to concentrated positive charge explains the rare large deflections.

3. Explain the stability problem

1 A circular orbit changes the direction of velocity, so the electron accelerates. Classical theory predicts that an accelerating charge radiates energy, which would make the electron spiral in.

2 Bohr proposed discrete stationary states. An electron in one such state keeps its energy. Energy is exchanged when it changes state.

4. Infer an energy change

1 Level 3 is higher in energy than level 2. The atom loses energy in the transition and emits light with energy equal to the difference.

5. Identify the missing particle

1 Two protons contribute about 2 u. The electrons add very little mass. The remaining mass of about 2 u is accounted for by two neutrons of roughly 1 u each.

6. Compare length scales

1 Divide the atomic diameter by the nuclear diameter. The metre units cancel.

$$\frac{10^{-10}}{10^{-15}} = 10^{5} = 100{,}000$$

2 The atomic diameter is about 100,000 times larger in this estimate.

Worked solutions • Counts, shells and isotopes

7. Decode an atom

1. Z = 16 identifies sulfur and gives 16 protons. The atom is neutral, so it has 16 electrons.
2. The neutron count is 34 − 16 = 18. Fill the first two shells with 2 and 8 electrons, leaving 6 in M. The configuration is 2, 8, 6.

8. Explain the difference

1. Silicon has four valence electrons and commonly forms four bonds, giving valency 4. Chlorine has seven valence electrons and needs one more to complete an octet, giving valency 1.

9. Classify a pair

1. The mass numbers are 18 + 22 = 40 and 20 + 20 = 40. The atomic numbers are different. These are argon-40 and calcium-40, a pair of isobars.

10. Calculate an average

1. Convert the abundances to 0.497 and 0.503. Multiply and add.

$$(0.497)(79\,\text{u}) + (0.503)(81\,\text{u}) = 80.006\,\text{u}$$

2. The result is just above 80 u because the heavier isotope is slightly more abundant.

11. Recover the abundance

1. Let f be the fraction of the heavier isotope. The lighter fraction is 1 − f.

$$10(1-f) + 11f = 10.8 \quad \Rightarrow \quad f = 0.8$$

2. The heavier isotope therefore makes up 80% of the sample.

12. Distinguish the numbers

1. A single nucleus contains a whole number of nucleons. The decimal is a weighted average of the masses of different isotopes in a sample.

Chapter review

Idea	What to remember
Identity	The proton count Z identifies the element.
Mass number	A counts protons and neutrons in one nucleus.
Neutral atom	Its proton and electron counts are equal.
Electron shells	The capacity is $2n^2$. Read the occupied shells from inside out.
Valency	Combining capacity. Use the outer electrons as a clue for simple compounds.
Isotopes	Same atomic number, different mass numbers.
Isobars	Different atomic numbers, same mass number.
Average atomic mass	Multiply each isotope’s mass by its fractional abundance and add.

Three checks before you answer

First decide whether the question is about the nucleus or the electrons. Next distinguish a count, such as mass number, from a mass measured in u. Finally, when comparing atoms, check Z before comparing A.

Reference

NCERT, **Exploration: Textbook of Science for Grade 9**, Chapter 8, Journey Inside the Atom, pp. 140–161. This Rama’s Academy edition follows the chapter’s topic sequence. Its explanations, worked questions and diagrams are independently written. The cover and laboratory scenes are generated illustrations. Colours and sizes in diagrams are schematic.